

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.1: Introduction to Chemistry — Teacher Reference (Pre-filled)

SUMMARY TABLE: CHEMISTRY GRADE 10	
Sub-Strand	Sub-Strand 1.1: Introduction to Chemistry
Driving Question	DRIVING QUESTION: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Kisumu Tray: Eight Substances, Many Puzzles	What did we observe about substances A-H on the Kisumu tray? Record the colour, form (powder, crystals, grains, liquid, tablet) and any other visible features of A table salt, B water, C sand, D sulphur powder, E iron filings, F antacid tablet, G paracetamol tablet and H sugar, and the sorting rules our group used.	What did we learn today? Write what we predicted for the water test and why, and what we realised about how looking alone may not tell us how a substance will behave. Also record the puzzling questions our group added to the Driving Question Board.	How can we begin to explain the phenomenon? In your own words, write your starting idea about why substances that look alike (for example salt, sugar and sand, or the antacid and paracetamol tablets) might behave differently, and one thing you will need to test in Lesson 2 to check your idea.
Lesson 2 Matter: Mixing, Separating and Heating the Kisumu Tray	Record what you observed for each of A-H in the water test and heat test, and the masses recovered from mixture M: iron 2.9 g, sand 2.8 g, salt 2.7 g, total 8.4 g, with a loss of 0.6 g from 9 g.	Write what you learned about properties of matter and states of matter: physical properties (solubility, magnetism, melting point, density) and chemical properties (burning, reacting with acid), and how solids, liquids and gases differ in their particles.	Explain in one or two sentences, using evidence, why substances that look like powders behave differently and why salt and sugar cannot be filtered out while sand and iron can be separated by filtration or magnetic separation, and how this links to the driving question about medicines, foods and safety.
Lesson 3 Matter Continued: Classifying Substances and Building Class Model v1	We observed that A salt and H sugar dissolve completely, C sand and E iron filings do not dissolve, E is pulled out by a magnet, D sulphur floats and burns with a blue flame and sharp smell, and F antacid fizzes in vinegar while G paracetamol does not. In mixture M we recovered iron 2.9 g, sand 2.8 g and salt 2.7 g, a total of 8.4 g out	We learned that substances can be classified as elements (D sulphur, E iron), compounds (A salt, B water, H sugar, calcium carbonate in F) or mixtures (M, and tablets F and G with binders). Mixtures can be separated by magnetic separation, filtration and evaporation, but dissolved salt needs evaporation. Class model v1 links	The substances behave differently because each has its own properties, which come from what it is made of. Iron and sand can be pulled out or filtered, while salt and sugar dissolve and cannot be filtered out. The antacid works because it neutralises acid, paracetamol works on pain and fever by a different action, and both are harmful

	of 9 g, so 0.6 g was lost.	composition to properties to uses and safety.	in excess, so we follow labels and correct dosage.
Lesson 4 Meaning of Chemistry as a Field of Science: Defining Chemistry from the Tray Evidence	What did you observe about the tray substances A-H that shows Chemistry studies matter? For example: salt A and sugar H dissolve, sand C and iron E do not, sulphur D burns with a blue flame, and the antacid F fizzes in vinegar while paracetamol G does not.	What did you learn about the meaning of Chemistry? Chemistry is the science that studies matter: its composition, properties, changes and uses. Matter is anything that has mass and occupies space.	How does the definition of Chemistry help explain why the antacid relieves heartburn while paracetamol relieves pain, and why chemists in Kenya need to know the properties of substances to make medicines, foods and detergents safely?
Lesson 5 Branches of Chemistry: Matching the Five Branches to the Kisumu Tray	We observed data from the Kisumu tray: sugar H and paracetamol G melt and brown at about 186 C and 169 C; salt A dissolves and iron E is pulled out by a magnet; sulphur D melts at about 115 C; mixture M (9 g) was separated by magnet (2.9 g), filtration (2.8 g) and evaporation (2.7 g); the antacid F fizzes in vinegar while paracetamol G does not.	Chemistry has five branches: organic, inorganic, physical, analytical and biochemistry. Each answers different questions about the same substances, and one substance can belong to more than one branch. Listening respectfully to every group member's opinion improved our answers.	The tray puzzles are answered by different branches working together: organic (sugar H and paracetamol G), inorganic (salt A and iron E), physical (melting points), analytical (separating mixture M) and biochemistry (how the antacid F and paracetamol G act in the body). This helps chemists make and test medicines, foods and other products safely, which moves us toward answering the driving question.
Lesson 6 Importance of Chemistry in Kenya: From the Kisumu Tray to Our Industries (Model v2)	Record what you observed: the properties of A-H (salt and sugar dissolve; sand and iron do not; iron is pulled by a magnet; sulphur melts and burns with a blue flame and sharp choking smell; F fizzes in dilute vinegar but G does not), and the Kenyan industry examples you studied.	Record what you learned: chemistry is important in agriculture, pharmaceutical, medicine, manufacturing, entertainment, nuclear, food, sports and energy industries because industries choose, separate and make materials using properties; the antacid neutralises acid, paracetamol relieves pain and fever, and the adult maximum is $8 \times 500 \text{ mg} = 4 \text{ g}$ in 24 hours.	Explain how this helps answer the driving question: use evidence from the tray and at least two industries to explain how properties help chemists make medicines, foods and other products, and how labels, correct dosage and avoiding substance misuse keep people safe. Describe what you added to model v2.
Lesson 7 Careers in Chemistry: Who Works with the Kisumu Tray Substances, and Who Gets to Choose?	We observed that careers such as pharmacist, analytical chemist, laboratory technician and quality control officer use the properties of substances we saw in the tray: F antacid reacts with acid, G paracetamol does not, and mixture M was separated by magnet, filtration and evaporation (2.9 g, 2.8 g and 2.7 g recovered out of 9 g).	We learned that chemistry offers many careers in medicine, industry and testing, and that gender stereotypes are fixed ideas that can wrongly limit who chooses these careers.	We can explain that people in chemistry careers use properties to make, separate and check products such as medicines, and that safety through labels and correct dosage matters, and that girls and boys can both choose these careers; this adds to our answer to the driving question.
Lesson 8 Consumer Protection and Product Labels: Reading the Story a Label Tells	We observed the labels on a paracetamol 500 mg box (G), an antacid pack (F), a maize flour packet, a juice carton and a detergent bottle. We found the KEBS Diamond Mark, expiry dates, ingredients and preservatives, nutritional information, and caution and dose messages, and we	We learned that labels protect consumers by giving information on quality, how long a product is safe to use, what it contains, and how to use it safely. The paracetamol label says adults take 1 to 2 tablets of 500 mg every 4 to 6 hours, with a maximum of 8 tablets ($4000 \text{ mg} = 4 \text{ g}$) in 24 hours,	Substances like the antacid (F) and paracetamol (G) have different properties and uses, so their labels give different instructions. Labels turn what chemists know about the properties of substances into safe-use messages for the consumer. This helps answer the driving question

	noticed where information was missing.	because more can damage the liver. Antacid taken too often can cause constipation or diarrhoea, and children must not be given adult doses.	about how we use products safely, and prepares us to study drugs, dosage and substance abuse in Lesson 9.
Lesson 9 Drugs and Substance Abuse: Why the Dose Makes the Difference	We observed that antacid F fizzes in dilute vinegar (acid) while paracetamol G does not; both are tablets that behave differently (F crumbles into a cloudy suspension in water, G stays mostly intact). We read from the label that the adult paracetamol maximum is 8 tablets of 500 mg in 24 hours, and that antacid taken too often can cause constipation or diarrhoea.	We learned that a drug is a substance that changes how the body works, a prescription is a health worker's written instruction for a medicine, dosage is the amount and timing, and substance abuse is using a substance in a harmful way or amount. $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ is the paracetamol maximum in 24 hours; more can damage the liver. Children must not be given adult doses.	The antacid relieves heartburn because it reacts with and neutralises excess stomach acid, while paracetamol relieves pain and fever by a different action and does not neutralise acid. Both are harmful in excess, so we follow the label, prescription and a health worker's advice, and we share this message in posters to keep our school and community safe.
Lesson 10 Safe and Healthy Learning Environment and the Final Model: How Properties of Substances Shape the Products and Medicines We Use	What we observed: Using the recorded results for A-H, write what you observed. For example, A salt and H sugar dissolve and cannot be filtered out; C sand and E iron filings do not dissolve, and E is pulled out by a magnet; D sulphur melts at about 115 °C then burns with a blue flame and a sharp choking smell; F antacid fizzes in vinegar but G paracetamol does not; mixture M gave 2.9 g iron, 2.8 g sand and 2.7 g salt (8.4 g recovered, 0.6 g lost).	What we learned: Write the ideas from Lessons 1-10 that help answer the driving question: each substance has its own properties which come from what it is made of; chemistry studies matter, its composition, properties, changes and uses; chemists use properties to choose, separate and make materials; labels (quality mark, expiry date, ingredients, caution messages), correct dosage and avoiding substance abuse keep people safe; learners have rights and responsibilities for a safe and healthy learning environment.	What we can now explain: In 8-10 sentences, explain how the different properties of substances help chemists make medicines, foods and other products, and how we use them safely. Include why the antacid relieves heartburn by neutralising acid while paracetamol relieves pain and fever by a different action, why both are harmful in excess (paracetamol maximum $8 \times 500 \text{ mg} = 4 \text{ g}$ in 24 hours), and one responsibility you will take for a safe and healthy environment at school and at home.

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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